

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 01
Sequential Circuit



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Recap of Previous Lecture



- Question Discussion on Comb. CKT
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Topics to be Covered

S-R Latch

S-R ff

[Sequential Circuit]

- Basic element of sequential circuit?

└→ S-R latch



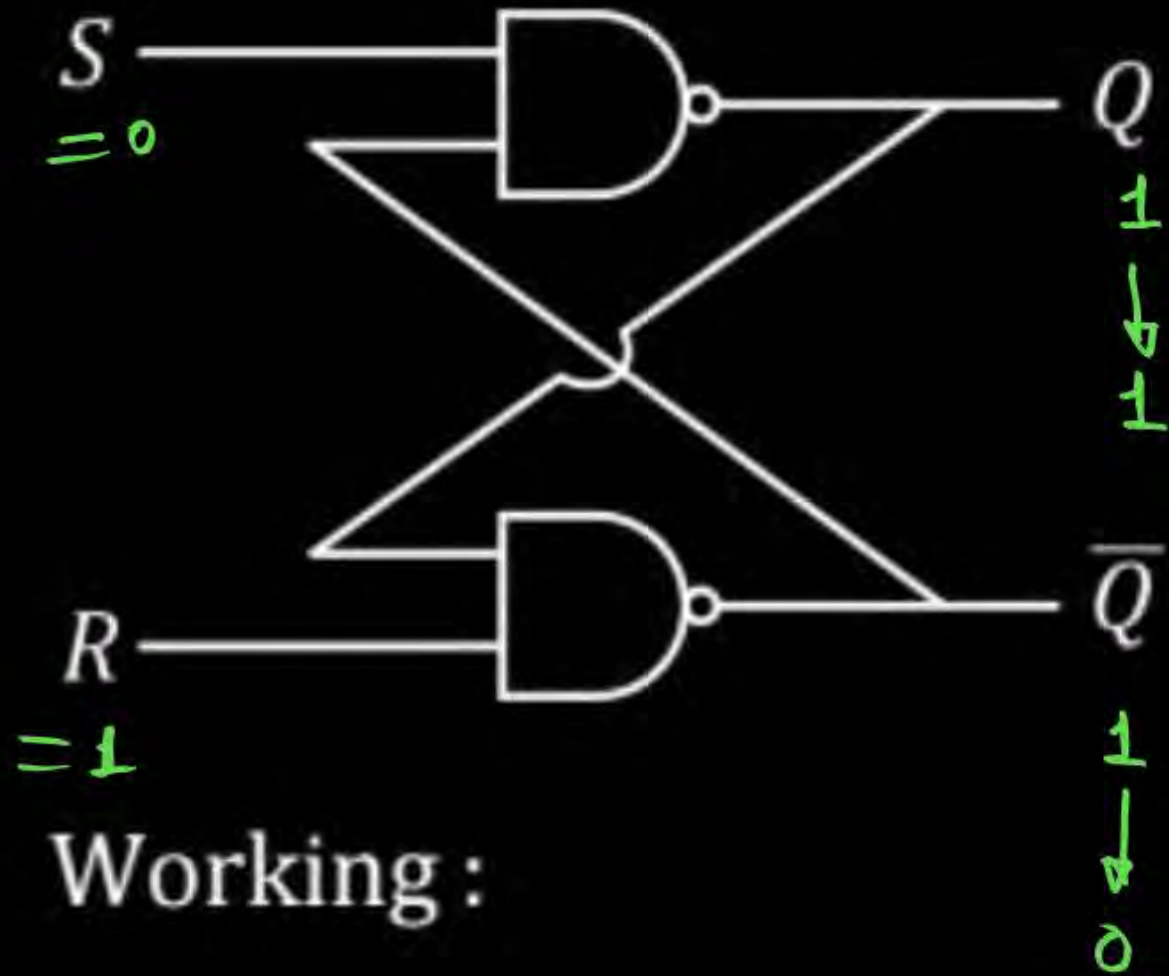
S-R ff



JK ff

S-R Latch :

SR latch using NAND gate :



→ Working :

S	R	$Q(n+1)$
0	0	invalid
0	1	1
1	0	0
1	1	$Q(n) \rightarrow \text{hold state}$

$$S=0, R=1$$

$$Q[n]=?$$

$$Q(n+1)=1$$

$$S=1, R=1$$

$$Q[n]=1$$

$$Q[n+1]=1$$

$$Q=1, R=0$$

$$Q[n]=1$$

$$Q[n+1]=0$$

$$S=1, R=1$$

$$Q[n]=0$$

$$Q[n+1]=0$$

$$S=0, R=0$$

$$Q[n]=0$$

$$Q[n+1]=1, \bar{Q}[n+1]=1$$

$$S=1, R=1$$

$$Q(n)=1, \bar{Q}(n)=1$$

in any of the valid o/p
state $(1,0)$ or $(0,1)$

$$S=0, R=0$$

$$(1,0) \text{ or } (0,1)$$

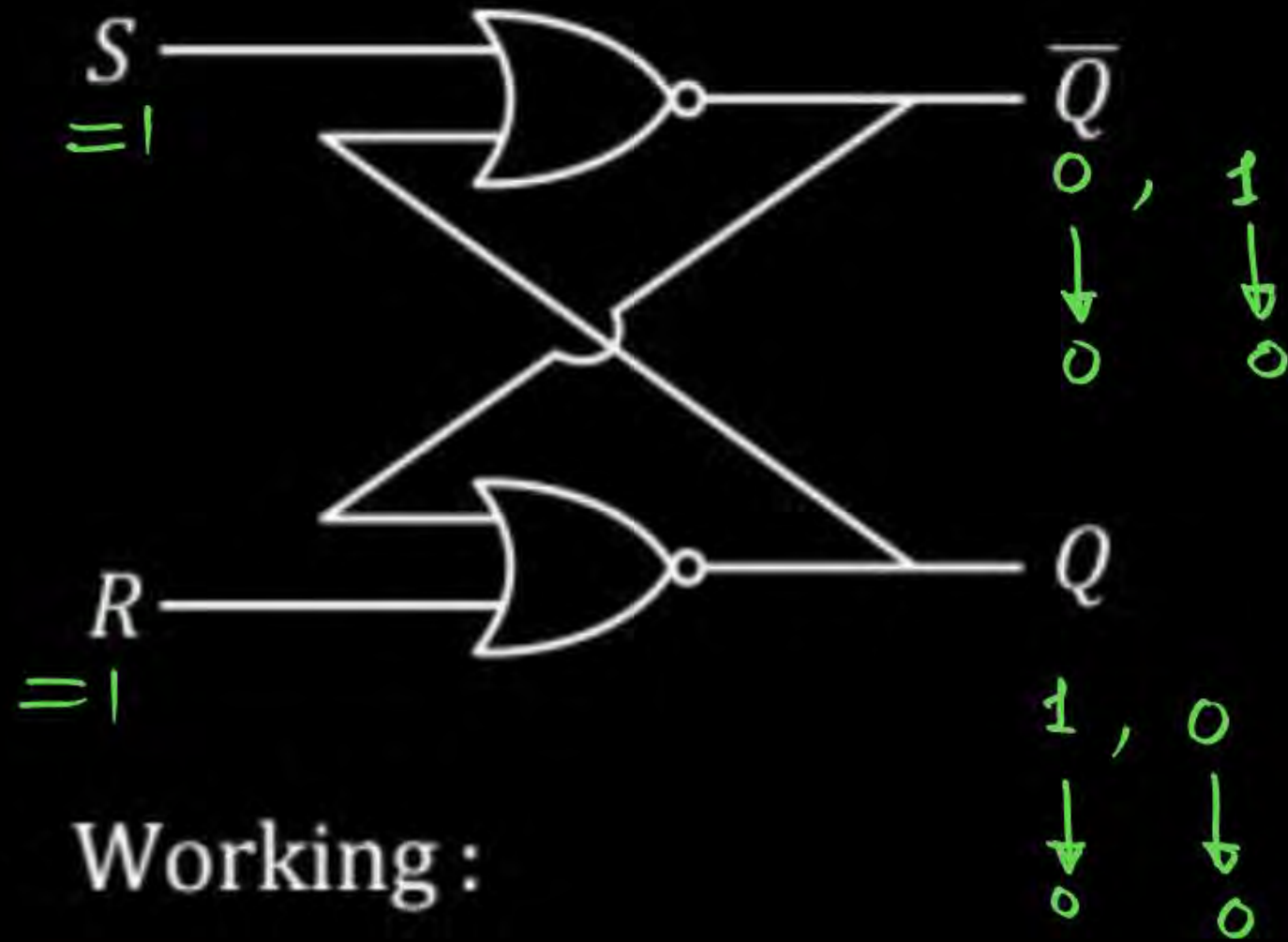
$$Q(n+1)=1, \bar{Q}(n+1)=1$$

$$S=0, R=1$$

$$(1,1)$$

$$Q(n+1)=1, \bar{Q}(n+1)=0$$

SR latch using NOR gate :



→ Working :

S	R	$Q(n+1)$
0	0	$Q(n)$
0	1	0
1	0	1
1	1	invalid state

[Question]

A NAND latch circuit is given below :



$S=0, R=0$ $W=?$ $W^+ = 1$
 $Z=?$ $Z^+ = 1$

$S=1, R=1$ $W=1$ $W^+ = 1, Z^+ = 0$
 $Z=1$ $W^+ = 0 \text{ or } Z^+ = 1$

Initially input $S = R = 0$ and after sometime input changes from $S = R = 0$ to $S = R = 1$ then

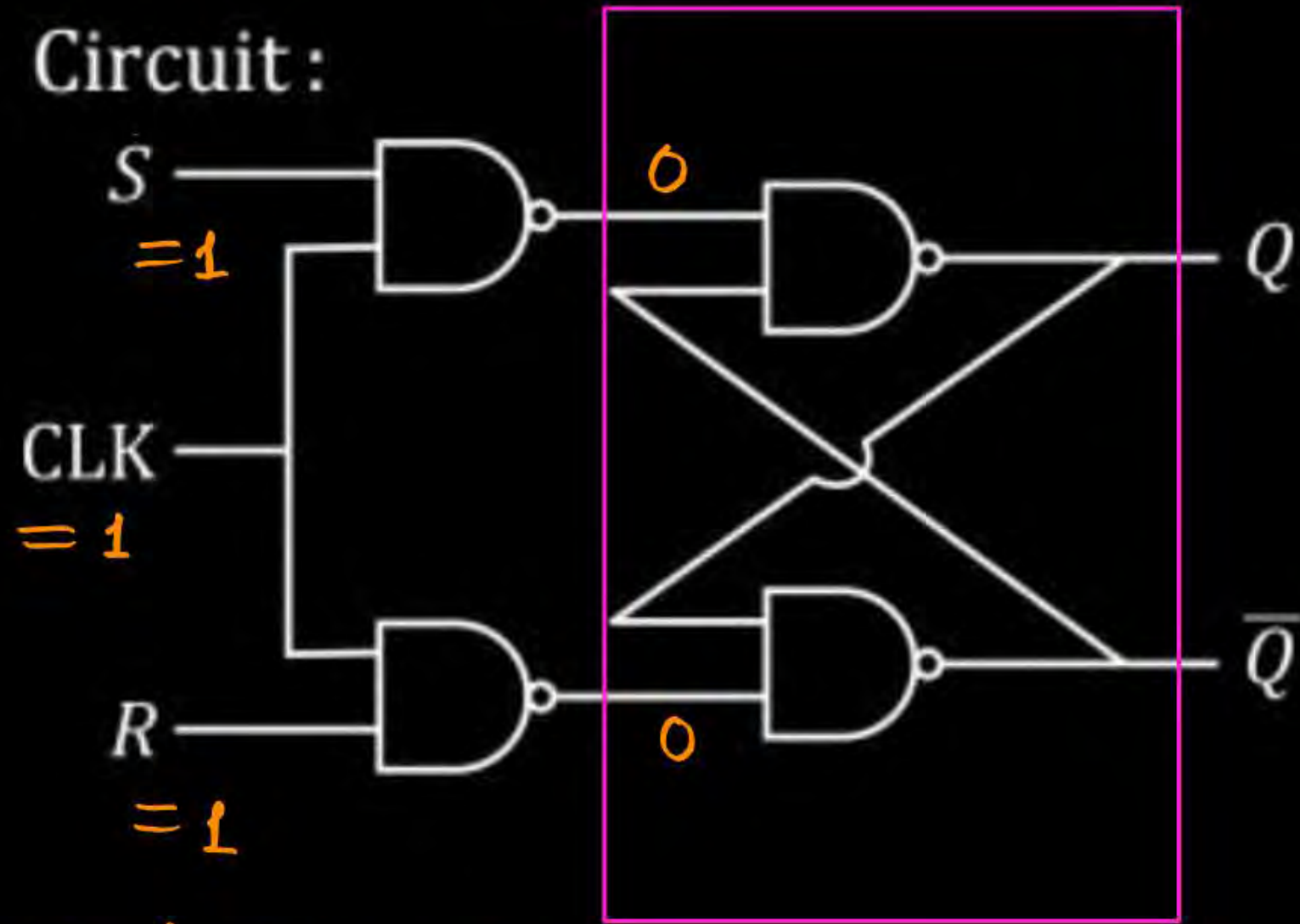
- (a) Output changes from $W = Z = 1$ to $W = Z = 1$. ~~X~~
- ☒ (b) Output changes from $W = Z = \underline{1}$ to either $W = 1$ and $Z = 0$ or $W = 0$ and $Z = 1$
- ~~X~~ (c) Output changes from $W = Z = 1$ to $W = Z = 0$ ~~X~~
- (d) None of these

Imp Point :

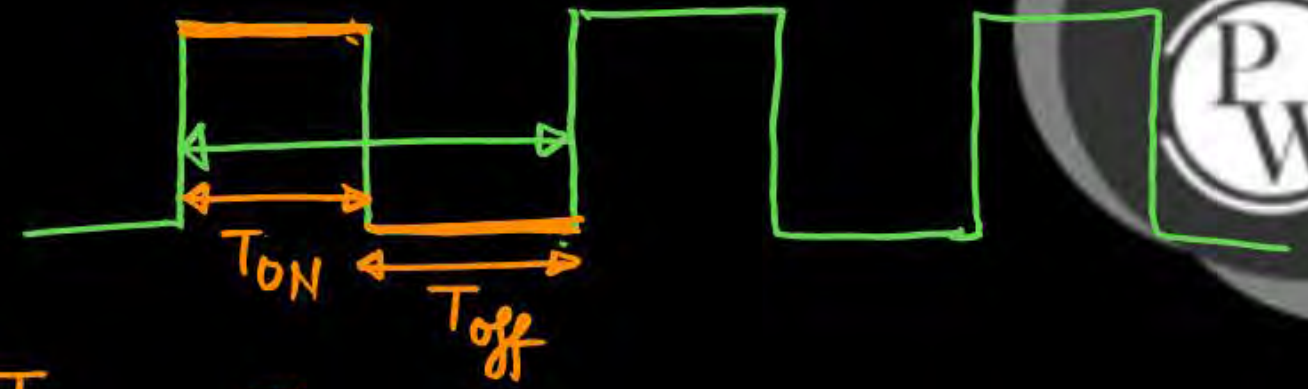
- Hold state i/p will hold the present o/p state in next o/p state only if present o/p state is a valid o/p state.
- If present o/p state is invalid o/p state then corresponding to hold state i/p, o/p will go to any of the valid o/p state $(1,0)$ or $(0,1)$ in next o/p state [does not hold the invalid o/p state].

[S-R Flip-Flop]

- Circuit:



+ve level triggered ff



$$T_{CLK} = T_{ON} + T_{off}$$

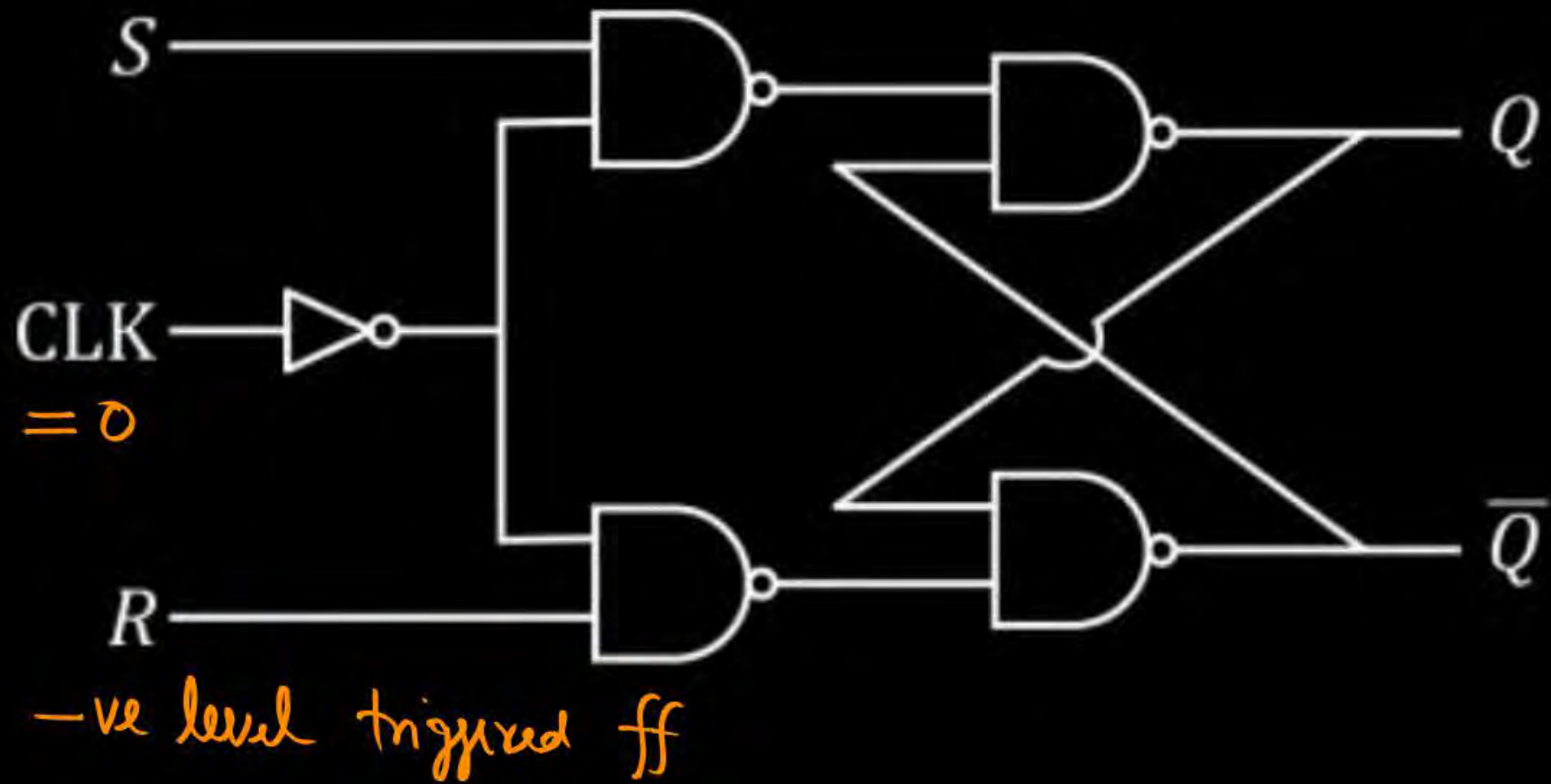
$$\% \text{ duty cycle} = \frac{T_{ON}}{T_{ON} + T_{off}} \times 100$$

$$= 50\% \text{ when } T_{ON} = T_{off}$$

$$T_{ON} > T_{off} \longrightarrow \text{duty cycle} > 50\%$$

$$T_{ON} < T_{off} \longrightarrow \text{duty cycle} < 50\%$$

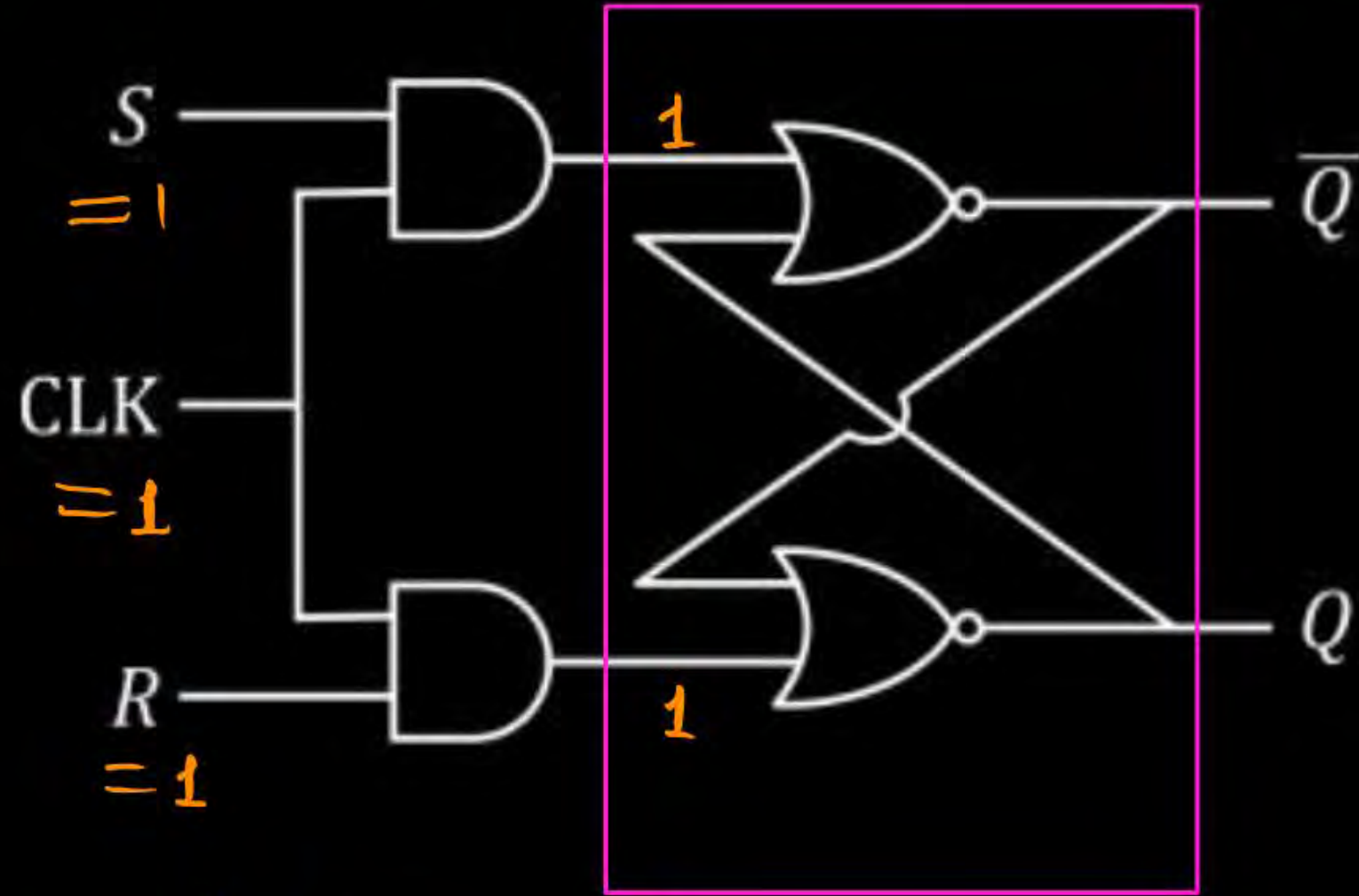
If circuit is changed to :



- Truth Table :

CLK	S	R	$Q(n+1)$
Inactive	X	X	$Q(n)$
Active	0	0	$Q(n)$ → hold state
Active	0	1	0
Active	1	0	1
Active	1	1	Invalid state

[S-R FF – using NOR latch + Clock]



+ve level triggered ff

Truth Table :

CLK	S	R	$Q(n+1)$
Inactive	X	X	$Q(n)$
Active	0	0	$Q(n)$
Active	0	1	0
Active	1	0	1
Active	1	1	Invalid

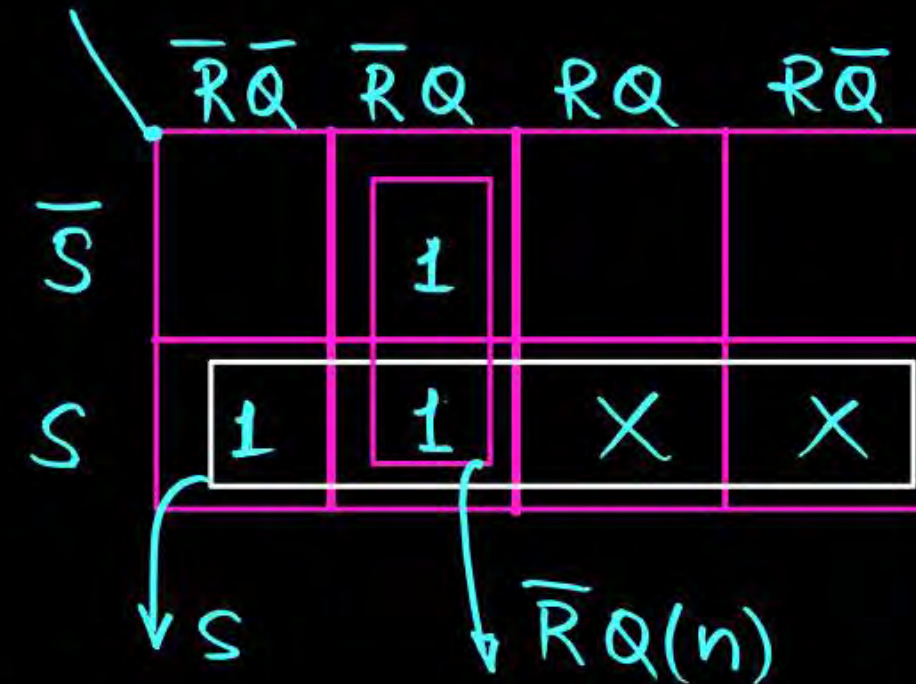
[Characteristic Table of SR Flip-Flop]



Clock Active

S	R	Q(n)	Q(n+1)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

$$Q(n+1) [S, R, Q(n)] = \Sigma(1, 4, 5) + d\Sigma(6, 7)$$



$$Q(n+1) = S + \bar{R}Q(n)$$

Characteristic equation

Valid only if $S \cdot R = 0$

[Excitation Table]

$Q(n)$	$Q(n+1)$	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

$$0 \xrightarrow[S=0, R=1]{S=0, R=0} 0$$

$$0 \xrightarrow{S=1, R=0} 1$$

$$1 \xrightarrow{S=0, R=1} 0$$

$$1 \xrightarrow[S=1, R=0]{S=0, R=0} 1$$



Topic : 2 Min Summary

- S-R latch
- S-R ff.

Thank you

GW
Soldiers !

